

CASE STUDY 2 - ANALYSIS OF ALGORITHM PARAMETERS

1. INTRODUCTION

In this report, the Differential Evolution algorithm (implemented using the ExpDE R package) was analysed to find the effect of changing its parameters. Furthermore, the following questions were answered:

1. Is there a difference in the mean performance of the algorithm when using these different recombination operators, for the given test problem?
2. In case there is, which is the best operator in terms of average performance (the lower the value returned, the better)?
3. and what is the magnitude of the differences found?
4. Is there an operator that should be recommended in the comparison of the others?

In order to find the answers to those questions the following experiments were performed.

2. EXPERIMENT PARAMETERS

The following experiments parameters are given for this study:

- Minimal difference of practical significance between any pair of algorithm (padronized as Cohen's d coefficient): $d^* = \delta^* / \sigma = 0.25$
- Desired significance: $\alpha = 0.05$
- Minimum power desired (for the case $d^* = d^*$): $\pi = 1 - \beta = 0.85$

The **R** language was used as main tool to perform the statistical tests. The ExpDE package was analysed in R to different groups of parameters that were pre-defined by the professor in the case study 2 handouts.

The fixed parameters for the algorithm (along the entire experiment) are *selpars*, *stopcrit* and *probpars*, as can be seen in Figure 1.

```
selpars <- list(name = "selection_standard")
stopcrit <- list(names = "stop_maxeval", maxevals = 60000, maxiter = 1000)
probpars <- list(name = "sphere", xmin = -seq(1,20), xmax = 20 + 5 * seq(5, 24))
```

Figure 1. Fixed parameters for the algorithm

The Figure 2 shows the recombination operators *recparsX*, *mutparsX*, *popsizex*, where X varies from 1 to 4. Those four operators will be studied in this report to figure out if they can improve or worsen the performance of the algorithm when they are changed.

```
## Operator 1 -- Arithmetic recombination
recpars1 <- list(name = "recombination_arith")
mutpars1 <- list(name = "mutation_rand", f = 4)
popsizex1 <- 300

## Operator 2 -- Binomial recombination
recpars2 <- list(name = "recombination_bin", cr = 0.7)
mutpars2 <- list(name = "mutation_best", f = 3)
popsizex2 <- 300

## Operator 3 -- BLX-alpha recombination
recpars3 <- list(name = "recombination_blxAlphaBeta", alpha = 0.4, beta = 0.4)
mutpars3 <- list(name = "mutation_rand", f = 4)
popsizex3 <- 230

## Operator 4 -- Eigenvector-based binomial recombination
recpars4 <- list(name = "recombination_eigen", othername = "recombination_bin", cr = 0.9)
mutpars4 <- list(name = "mutation_best", f = 2.8)
popsizex4 <- 85
```

Figure 2. Recombination operators of the algorithm

To obtain one observation of the performance of the algorithm with a given operator, the commands in Figure 3 were used. Where *recparsX*, *mutparsX*, *popsizex* are the parameters mentioned previously.

```
out <- ExpDE(popsizex, mutparsX, recparsX, selpars, stopcrit, probpars)
out$Fbest
```

Figure 3. Commands to obtain one observation of the performance of the algorithm

3. STATISTICAL ANALYSIS AND RESULTS

In this section some statistical experiments will be performed to compare the operator and their influence on the performance of the Differential Evolution algorithm.

3.1. DEFINITION OF THE HYPOTHESIS

In this experiment, the null hypothesis and the alternative hypothesis are given, as in the ANOVA test, in the two equations below. The null hypothesis assumes that the effect of the i-th level is equal to zero, which meaning that there is no set of parameters that has a

better performance than the others. The alternative hypothesis assumes that there is at least two groups that have different performance from each other.

$$H_0 : \tau_i = 0, i = (1, 2, 3, 4)$$

$$H_1 : \exists \tau_i \neq 0$$

3.2. ESTIMATION OF THE POPULATION SIZE

For this estimation, the power t-test was used, however, the variable δ was not specified in the experiment. To find δ , 50 observations of each one of the 4 operators was extracted. After the extraction, the standard deviation of each one of them was calculated, as below. The variable resultX (X=1~4) is a vector of 50 observations for each of the 4 operators.

```
std1 <- sd(result1)
std2 <- sd(result2)
std3 <- sd(result3)
std4 <- sd(result4)
```

After estimating the standard deviation of the 50 samples, the Cohen's d coefficient d^* was used to calculate δ and, finally, find the size of the population. The equation below was used in the power t-test, where σ is std1, std2, std3 and std4.

$$d^* = \delta^* / \sigma = 0.25$$

$$\delta = \sigma * 0.25$$

The Figure 4 shows the power t-test for the first operator, using a power of 0.85, a significance level of 0.05 and a two sided one sample test. This estimation gives as result the size of the population, in this case 146 (round up 145.5876) observations are needed to achieve such requirements of power and standard deviation.

```
> power.t.test(power = 0.85, delta = std1*0.25, sd = std1, sig.level = 0.05,
+             type = "one.sample", alternative = "two.sided")
```

```
One-sample t test power calculation
```

```
      n = 145.5876
  delta = 0.7276307
      sd = 2.910523
sig.level = 0.05
  power = 0.85
alternative = two.sided
```

Figure 4. Estimation of the population for the first operator

It was interesting to notice that even though the standard deviations of the samples are different, the population has exactly the same size for each one of the parameters. As can be seen in Figures 5, 6 and 7, the same size of 146 observations was achieved for different parameters.

```
> power.t.test(power = 0.85, delta = std2*0.25, sd = std2, sig.level = 0.05,
+             type = "one.sample", alternative = "two.sided")
```

One-sample t test power calculation

```
      n = 145.5876
    delta = 1.027274
      sd = 4.109095
sig.level = 0.05
  power = 0.85
alternative = two.sided
```

Figure 5. Estimation of the population for the second operator

```
> power.t.test(power = 0.85, delta = std3*0.25, sd = std3, sig.level = 0.05,
+             type = "one.sample", alternative = "two.sided")
```

One-sample t test power calculation

```
      n = 145.5876
    delta = 0.828078
      sd = 3.312312
sig.level = 0.05
  power = 0.85
alternative = two.sided
```

Figure 6. Estimation of the population for the third operator

```
> power.t.test(power = 0.85, delta = std4*0.25, sd = std4, sig.level = 0.05,
+             type = "one.sample", alternative = "two.sided")
```

One-sample t test power calculation

```
      n = 145.5876
    delta = 1.434315
      sd = 5.737261
sig.level = 0.05
  power = 0.85
alternative = two.sided
```

Figure 7. estimation of the population for the fourth operator

3.3. STATISTICAL ANALYSIS AND HYPOTHESIS TESTING

After collecting the population of 146 samples of each one of the operators, the ENOVA test was performed. This test is more suitable for this situation because we are interested on the comparison of the four groups of parameters.

The ANOVA table, in the Figure 8, provides information on the sources of variation, together with their corresponding d.o.f., sums of squares and mean square values. The table

also informs the values of the test statistic and the corresponding p-value of the test (Pr (> F)).

```
> result <- aov(MyData$f_best ~ MyData$operator, data=MyData)
> summary.aov(result)
              Df Sum Sq Mean Sq F value    Pr(>F)
MyData$operator  1    845   844.8    7.413 0.00667 **
Residuals       582  66326   114.0
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 8. ANOVA table of test results and estimations

In this experiment, the p-value ($p = 0.00667$) suggests the **rejection of the null hypothesis** in favor of the alternative hypothesis. It means that at least one of the operators has an effect significantly different from zero.

3.4. VERIFICATION OF THE TEST ASSUMPTION

The ANOVA model is based on three assumptions on the behavior of the residuals. In this section, the assumptions below will be tested.

- Homoscedasticity, i.e., equality of variances across groups;
- Normality
- Independence;

First, the **normality** can be tested using the Shapiro-Wilk test and a normal qqplot of the residuals. The hypothesis for this test is given in the equations below:

$$H_0 : \text{data is normal}$$

$$H_1 : \text{data is not normal}$$

Performing the Shapiro-Wilk test, we obtain a p-value of $2.2 \cdot 10^{-16}$, which means that we have evidence enough to reject the null hypothesis.

```
> shapiro.test(result$residuals)
      Shapiro-Wilk normality test

data:  result$residuals
W = 0.91247, p-value < 2.2e-16
```

Furthermore, after plotting the data using qqplot along with the shapiro-test (Figure 8), we have data the data is not inside the expected values for normality, therefore we can reject the null hypothesis again.

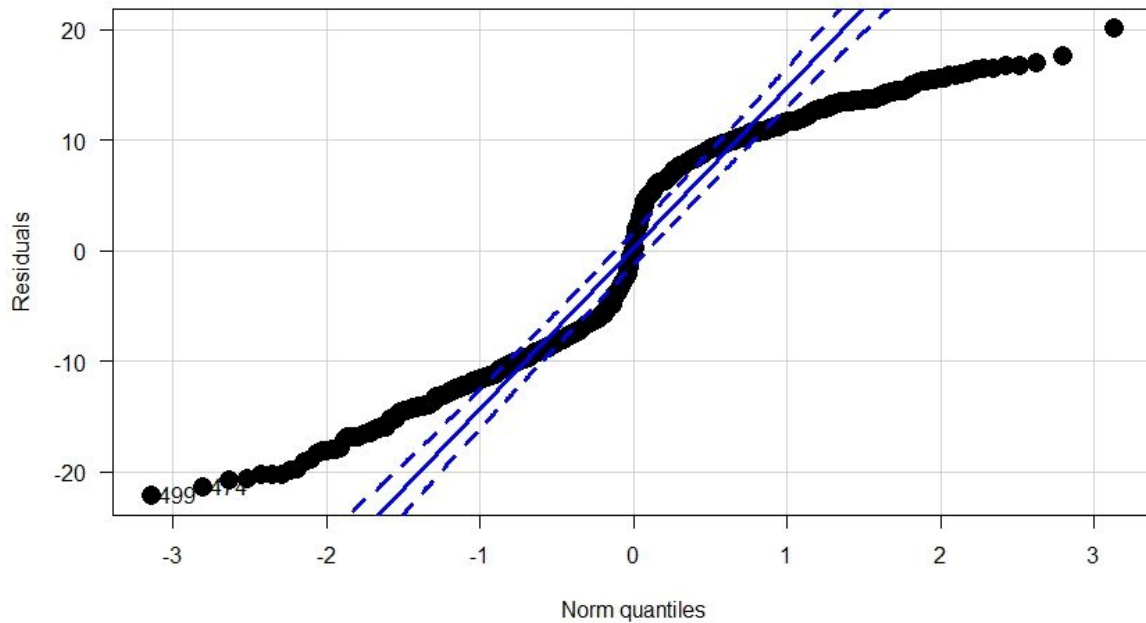


Figura 8. Graphic of Shapiro-Wilk test coupled with qqplot

Next, the **homoscedasticity** assumption can be verified by the Fligner-Killeen test, together with plots of residuals by fitted values. The hypothesis for this test is given in the equations below:

H_0 : the variance of the data is homogeneous

H_1 : the variance of the data is not homogeneous

After executing the command below, the p-value was also of $2.2 \cdot 10^{-16}$, which leads to the rejection of the null hypothesis, because the variances of the data are not homogeneous.

```
> fligner.test(MyData$f_best ~ MyData$operator, data = MyData)

Fligner-killeen test of homogeneity of variances

data:  MyData$f_best by MyData$operator
Fligner-killeen:med chi-squared = 82.456, df = 3, p-value < 2.2e-16
```

The Figure 9 shows the plot of the residuals by fitted values. This shows that there are some sets of operator in which the variance is close from each other, however the variance is not homogeneous for other pairs of parameters.

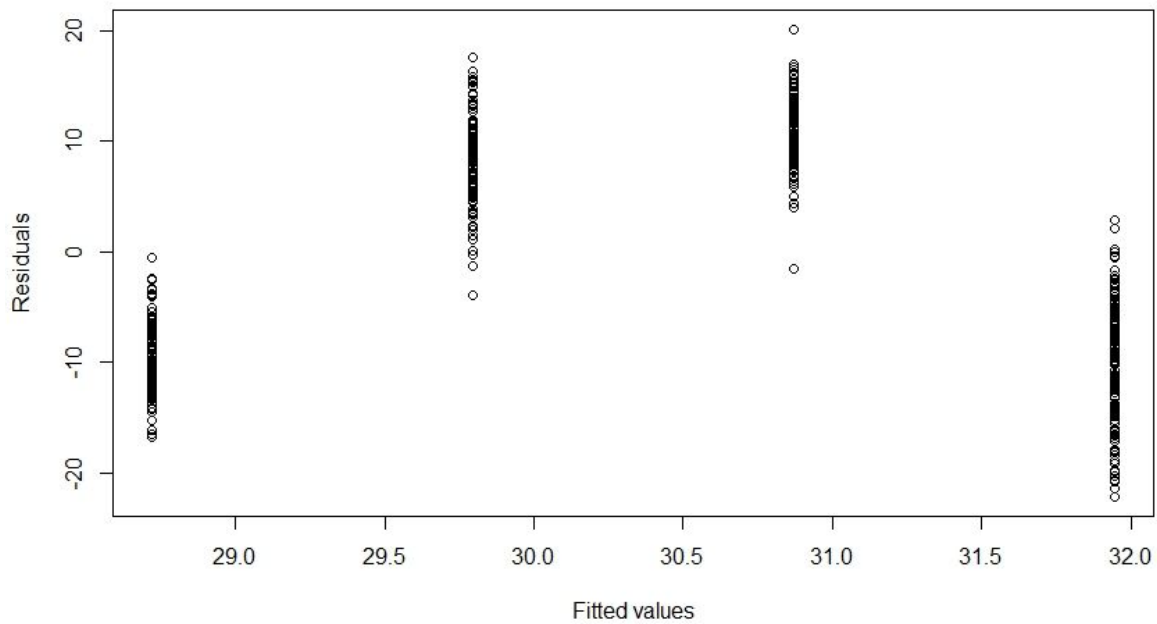


Figure 9. Test of homogeneity of variances

Lastly, the **independence** assumption was guaranteed on the project design phase, as well as on the analysis. This includes avoiding pseudoreplication and ordering effects, among others. In this project, the data is independent from each other.

3.4. ESTIMATION OF THE EFFECT SIZE AND THE CONFIDENCE INTERVALS

For these two estimations, cohen's d function was used as a method to find the means and the confidence level of each one of the distributions. The table 1 shows the result for all the possible combinations of operators taken two by two.

Pair	Effect Size	Confidence level	
		Lower	Upper
1,2	-5.702697	-6.221135	-5.184260
1,3	-7.628424	-8.291042	-6.965805
1,4	-0.5299199	-0.7642859	-0.2955540
2,1	5.702697	5.184260	6.221135
2,3	-1.073538	-1.319930	-0.8271452
2,4	3.503921	3.137176	3.870667
3,1	7.628424	6.965805	8.291042
3,2	1.073538	0.8271452	1.3199301
3,4	4.50469	4.071487	4.937893
4,1	0.5299199	0.2955540	0.7642859
4,2	-3.503921	-3.870667	-3.137176
4,3	-4.50469	-4.937893	-4.071487

Table 1. Effect size and Confidence interval of the operators

The Cohen's function calculated the difference between the means of two operators. Therefore we can conclude that the smallest mean is the mean of the performance of the first operator, because the mean of operator 1 is smaller than the means of operators 2, 3 and 4, for this reason the effective size is negative. The second best operator is the operator 4, because its mean is smaller than the mean of the operators 2 and 3, but it is higher than the mean of operator 1. The third best is the operator 2, because its mean is smaller than the operator 3, for this reason it is negative. The lower performance came from the operator 3.

Operator	Mean
1- Arithmetic recombination	19.29935
2- Binomial recombination	38.36047
3- BLX-alpha recombination	42.0036
4- Eigenvector-based binomial recombination	21.67094

Table 2. Mean of the performance of the operators

The table 2 shows that the operator 1 which used Arithmetic recombination had the **best** performance with mean equals to 19.29935, while the **worst** operator was the one which used BLX-alpha recombination with mean equals to 42.0036.

4. CONCLUSION

In this report, it was asked to compare the performance of four parameters using the statistical tests. It was observed in the results that the operators had different performances, because they solved the same problem using different approaches.

The ANOVA test was used as the main statistical inference method. As there are 4 sets of parameters ANOVA can be applied. However, an interesting aspect of the collected data is that it is not normal and the variances are not homogeneous, which goes against 2 of the 3 assumptions of the ANOVA test. Nevertheless, the degree of surprise (p-value) from ANOVA gave evidence enough to reject the null hypothesis.

Even though Anova had rejected the null hypotheses, there is not enough results to express which one of the operators had better or worse performance. For this reason, other methods had to be applied to calculate the difference between the means, confidence level and the means for each operator.

From the results above, this report concludes that the best set of parameters was observed in the operator 1, which implements Arithmetic recombination to solve a problem in Differential Evolution algorithm, the operator 1 had a mean of 19.299 in its performance, while the operator 3, which implemented BLX-alpha recombination, had the worst performance with mean 42.0036.

5. FINAL CONSIDERATIONS

This experiment has some limitations that comes from the assumptions of the ANOVA test. Even though the data does not follow a normal distribution or the variances are not homogeneous, the ANOVA test was applied. In my opinion, the assumptions should be verified before running the ANOVA test, therefore if the data is out of any of the assumptions, the student should apply any method to treat the data before applying the test.